

Exploratory Data Analysis: Patient Lifestyle & Disease Risk

Project: Data Analytics Internship – Project 2 (EDA), Decode Labs

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Dataset: 504 patient records – demographics, lifestyle habits, and lab values

1. Problem Statement

Can lifestyle and lab data – age, BMI, sleep, exercise, smoking, alcohol use, cholesterol, blood pressure, and glucose – explain variation in a patient's overall disease risk score? Before any predictive modeling, this report audits the data for quality issues and surfaces the patterns, trends, and outliers hidden inside it.

2. Methodology

1. Loaded the raw dataset and audited it for duplicates and missing values (the "forensic evidence" pass).
2. Removed duplicate records and calculated a five-number summary (min, Q1, median, Q3, max) for every numeric column.
3. Flagged outliers using the IQR method ($< Q1 - 1.5 \times IQR$ or $> Q3 + 1.5 \times IQR$) and reviewed each flagged group

individually — deciding what was noise (entry error) vs. signal (genuine extreme case).

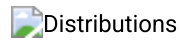
4. Ran a Pearson correlation analysis across all numeric variables against `disease_risk_score`.
5. Broke risk down by smoking status as a categorical comparison.

3. Data Quality Audit

Check	Result
Duplicate rows found & removed	4
Overall data completeness (post-dedup)	98.0%
Columns with missing data	<code>bmi</code> , <code>sleep_hours</code> , <code>alcohol_units_per_week</code> , <code>cholesterol</code> — each ~6% missing

The missingness is spread evenly across four columns at a consistent ~6% rate, which suggests a systematic collection gap (e.g., a form field skipped in a subset of intake sessions) rather than random data loss. This is worth flagging to whoever owns the intake process, not just imputing silently.

4. Distributions — The Geometry of the Data



Key observations:

- `age`, `bmi`, `sleep_hours`, `bp_systolic`, `glucose_level`, and `disease_risk_score` are roughly symmetric – **mean and median are close and both are trustworthy.**
- `exercise_hours_per_week` (mean 2.4 vs. median 1.7) and `alcohol_units_per_week` (mean 3.0 vs. median 2.1) are **right-skewed** – most patients exercise/drink lightly, with a long tail of high-activity or high-consumption outliers pulling the mean upward. **Median is the more honest "typical patient" figure for these two.**

5. Outlier Detection – Boxplots (IQR Method)



Variable	Outliers flagged	Verdict
<code>cholesterol</code>	7 (incl. 3 above 400)	Noise. Values of 520, 495, 480 are physiologically implausible for this metric on this scale – almost certainly entry errors (likely a

Variable	Outliers flagged	Verdict
		misplaced digit). Recommend flagging for re-entry, not deletion.
bp_systolic	7 (incl. 3 at 205–215)	Signal. These are high but medically plausible (hypertensive crisis range) – worth clinical follow-up, not removal.
bmi	3	Mixed. One value of 3.2 is an obvious entry error (likely meant 32.0); 61.5 and 58.2 are extreme but plausible.
exercise_hours_per_week , alcohol_units_per_week	28, 18	Expected – these are the right-skewed variables; IQR naturally flags the long tail.

Variable	Outliers flagged	Verdict
		Not data errors.

Takeaway: roughly 1.2% of records contain values needing manual review before this dataset would be ready for modeling — a routine, not alarming, cleanup step.

6. Correlation Analysis

 Correlation Heatmap

Strongest correlations with `disease_risk_score` :

Variable	r
age	0.56
bp_systolic	0.51
cholesterol	0.42
bmi	0.39
glucose_level	0.24
exercise_hours_per_week	-0.10

The Golden Rule check: age correlates with bp_systolic (0.58), cholesterol (0.37), and bmi (0.14) simultaneously — meaning age is likely acting as a **confounder** behind several of these

relationships rather than each variable independently driving risk. A model built on this data should control for age before crediting BMI or cholesterol with independent predictive power.

BMI vs Risk Scatter

The scatter plot also shows exercise hours (color) clustering toward the lower-BMI, lower-risk region — consistent with exercise's negative correlation with risk, even though the correlation coefficient itself (-0.10) is modest.

7. Risk by Smoking Status

Risk by Smoking

Smoking status	Mean risk	Median risk	n
Non-smoker	51.4	52.0	296
Former smoker	57.6	58.0	130
Smoker	65.9	64.7	74

Smokers show a **~14.5 point higher average risk score** than non-smokers — the single clearest categorical split in the dataset.

8. Synthesizing the Case — "So What?"

What the data says	The diagnosis
Age, BP, cholesterol, and BMI are the top four risk correlates	A future risk model should prioritize these four fields, but must control for age to isolate their independent effects
Smokers average 65.9 vs. 51.4 for non-smokers	Smoking status alone separates the highest-risk cohort — a natural first filter for any intervention program
Exercise and alcohol are right-skewed	Reporting a "typical" patient's activity level should use the median, not the mean, or population outreach messaging will be miscalibrated
~6% missing in four fields, systematic not random	Worth a conversation with data collection owners before this dataset is trusted for downstream modeling

9. Recommendations

1. Fix the three cholesterol entry errors and the one BMI entry error before any modeling — they materially distort the mean.
 2. Investigate the systematic ~6% missingness pattern in intake data collection.
 3. For a follow-on predictive model (Project 3+), control for age as a confounder rather than treating BP/cholesterol/BMI as fully independent predictors.
 4. Smoking status is the strongest, cleanest categorical lever — a good candidate for a first-pass risk stratification rule.
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*Files accompanying this report: raw dataset
(`healthcare_lifestyle_data.csv`), cleaned dataset
(`healthcare_lifestyle_data_cleaned.csv`), and the full
analysis script (`eda_analysis.py`).*